1.3 - Wake Functions

- The ME for the fields generated by a source particle of transv. coord.'s (x_0, y_0) and velocity v along the longitudinal direction s

  \+

- Boundary Conditions of continuity for $D^{\perp}$, $B^{\perp}$, $\vec{E}^{\parallel}$ and $\vec{H}^{\parallel}$

  \+

- Expression for momentum change of the witness particle (determ. using the mentioned approximations)

$\Rightarrow$ Fields $\vec{E}$ and $\vec{B}$ from the passage of the source part. are uniquely determined

$$\Delta\vec{p}(\vec{\rho}_s, \vec{\rho}_w, z) = \int_{-\infty}^{\infty} dt\, \vec{F}(\vec{\rho}_s, \vec{\rho}_w, s, t)\Big|_{s = vt - z} \quad ;$$

$\vec{E} = \vec{E}(\vec{\rho}_w, vt-z)$

$\vec{B} = \vec{B}(\vec{\rho}_w, vt-z)$

Lorentz Force

where: $\vec{\rho}_s = (x_s, y_s)$, $\vec{\rho}_w = (x_w, y_w)$, $\vec{F} = q\left[E_s\,\hat{s} + (E_x + vB_y)\,\hat{x} + (E_y - vB_x)\,\hat{y}\right]$

In theory

$\Rightarrow$ Everything we need to compute the total momentum variation of each witness particle due to the effect of all other (source) particles !

<u>Nonetheless</u>, we can use some <u>additional definitions</u> that will be very handy when one seeks to indeed perform such calculations (and explore symmetries / simplifications).

The first of them are the <u>Wake functions</u> (or <u>wakes</u>):

$W_s > 0 \Rightarrow$ energy loss *

$$(W_x, W_y, W_{\parallel}) = \frac{v}{qQ}(\Delta p_x, \Delta p_y, -\Delta p_s) \sim [V/c] = [J/c^2]$$

$\underbrace{p \cdot v}_{\sim m v^2\,[kg\,m^2/s^2]} \sim \underbrace{F \cdot d}_{\sim m \cdot a \cdot d\,[kg\,m^2/s^2]}$

functions <u>independent</u> of the charges involved !

$q_w \sim q$

$q_s \sim Q$

$\sim$ W_i's depend only on the "structure" that generates the wake fields

* This section is mainly based on [de Sá, PhD Thesis, 2018] and [Stupakov, 2000]

1.4 - Wake Potential

Then, having the wake functions $\vec{w}$, one elegant (and powerful) further step is showing that these functions can be obtained from the derivatives of a scalar potential function W, which we'll call wake potential

REMINDER

In eletromag: $\vec{E} \rightarrow$ irrotational field $\Rightarrow \oint \vec{E}\cdot\vec{dl} = 0 \Rightarrow \int_A^B \vec{E}\cdot\vec{dl}$ is path independent

$\rightarrow$ enables defining $V(\vec{r}) \triangleq -\int_G^{\vec{r}} \vec{E}\cdot\vec{dl} \Rightarrow -\int_A^B \vec{E}\cdot\vec{dl} = V_B - V_A$

$$\Rightarrow \quad \vec{E}(\vec{r}) = -\vec{\nabla}\cdot V(\vec{r}) \quad \text{or simply} \quad \vec{E} = -\vec{\nabla}V$$

$\hookrightarrow \sim E_u(\vec{r}) = -\frac{\partial V}{\partial u}(\vec{r})$; $\vec{E}$: vector field ; $-\vec{\nabla}V$: gradient of a scalar potential

And just noticing:

$$E_u = \frac{F_u}{q} = \frac{1}{q}\cdot\frac{dp_u}{dt} \quad ; \quad w_u \sim \frac{V}{qQ}(\pm)\, dp_u$$

We want to find a $W(\vec{\rho}_s, \vec{\rho}_w, z)$ such that $w_u \propto \frac{\partial W}{\partial u}$

In particular, we'll have:

$-\frac{\partial W}{\partial s} \sim$ (the minus sign being due to $w_{\parallel} = -C\cdot\Delta p_s$, $C>0$)

$$\boxed{w_{\parallel} = -\frac{\partial W}{\partial(-z)} = \frac{\partial W}{\partial z} \quad ; \quad \vec{w}_{\perp} = \vec{\nabla}_{w,\perp}\cdot W = \left(\frac{\partial}{\partial x_w}, \frac{\partial}{\partial y_w}\right)\cdot W}$$

To find such potential, we start by writing the Lagrangian of the problem and, from the Euler-Lagrange equation $\left(\frac{d}{dt}\left(\frac{\partial L}{\partial \vec{v}}\right) = \vec{\nabla}L\right)$, obtain an expression for $\vec{\Delta p}$ ("$\propto$" $\vec{w}$), which will, in this specific case, be proportional to the gradient of a scalar function!

$\rightarrow$ the " " are due to the minus sign in $w_{\parallel} = -\Delta p_s$, so $\vec{w} \neq a\cdot\vec{\Delta p}$, meaning $\vec{w}$ and $\vec{\Delta p}$ are not literally proportional

APPENDIX

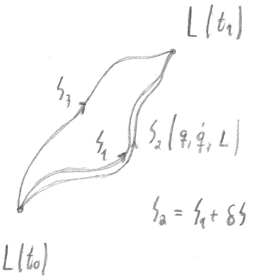
Reminder on Lagrangian Mechanics

definition in classical mechanics

- We begin with the Lagrangian: $L = T - V$

$L = L(q, \dot{q}, t)$

- From which we can define the action: $S = \int L \, dt$



- Then, from the Principle of Least Action, we

know that any system will evolve in a way which extremizes $S \sim \delta S = 0$!

- But what does $\delta S = 0$ mean? Say we begin at the optimal path $(q, \dot{q})$, so that S_i is the "extreme-valued" action. Then, when making a small deviation $(\delta q, \delta \dot{q})$ from that path, we should find $S_f = S_i + \delta S = S_i$,

i.e., $\delta S = 0$:

$\delta \dot{q} = \delta \frac{dq}{dt} = \frac{d}{dt} \delta q$

$$S(q + \delta q, \dot{q} + \delta \dot{q}) = \int_{t_0}^{t_1} L(q + \delta q, \dot{q} + \delta \dot{q}, t)\, dt = \int_{t_0}^{t_1} \left[L(q, \dot{q}, t) + \frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right] dt =$$

1st order $\sim \delta$'s are small

$$= S(q, \dot{q}, t) + \int_{t_0}^{t_1} \frac{\partial L}{\partial q} \delta q \, dt + \int_{t_0}^{t_1} \underbrace{\frac{\partial L}{\partial \dot{q}}}_{u} \underbrace{\frac{d}{dt} \delta q}_{dv/dt} \, dt$$

$\delta \dot{q} = \dot{q}_f - \dot{q}_i = \frac{d}{dt}(q_f - q_i) = \frac{d}{dt} \delta q$

$\delta q(t_0) = \delta q(t_1) = 0$

$$\rightarrow = \cancel{\left. \underbrace{\frac{\partial L}{\partial \dot{q}}}_{u} \underbrace{\delta q}_{v} \right|_{t_0}^{t_1}} - \int_{t_0}^{t_1} \underbrace{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}} \right)}_{du/dt} \cdot \underbrace{\delta q}_{v} \, dt$$

$$\Rightarrow \delta S = S(q + \delta q, \dot{q} + \delta \dot{q}) - S(q, \dot{q}, t) = \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}} \right) \right] \delta q \, dt$$

- If instead of $L(q, \dot{q}, t)$ we had $L(q_i, \dot{q}_i, t)$, we would have $\delta L = \sum_i \frac{\partial L}{\partial q_i} \delta q_i + \sum_i \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i$, and thus

$$\delta S = \int_{t_0}^{t_1} \delta L(q_i, \dot{q}_i, t)\, dt = \sum_i \left[\int_{t_0}^{t_1} \frac{\partial L}{\partial q_i} \delta q_i \, dt + \int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i \, dt \right] = \int_{t_0}^{t_1} \sum_i \left[\left(\frac{\partial L}{\partial q_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} \right) \right) \delta q_i \right] dt = 0$$

Now, to have $\delta S = 0$ for any small δq_i's, the remaining of the integrands must all be zero everywhere, and thus
↳ independently

Euler-Lagrange equation *

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0} \quad \forall i$$

• Thus, by building the Lagrangian $L(q_i, \dot{q}_i, t)$ of a physical system, we can determine how $(q_i, \dot{q}_i)$'s will evolve – i.e., the equations of movement of the system – by using $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}$.

• Illustrating the case for classical mechanics (1 particle):

▸ $T = \frac{1}{2} m |\vec{\dot{q}}|^2 \Rightarrow L(\vec{q}, \vec{\dot{q}}, t) = \frac{1}{2} m \underbrace{|\vec{\dot{q}}|^2}_{= \frac{1}{2}m(\dot{q}_x^2 + \dot{q}_y^2 + \dot{q}_z^2)} - V(\vec{q})$

Each coord. v:

▸ Euler – Lagrange: $\frac{\partial L}{\partial \dot{q}_v} = m\dot{q}_v \Rightarrow \frac{d}{dt}(m\dot{q}_v) = \frac{\partial}{\partial q_v}\left(-V(\vec{q})\right) \Rightarrow m\ddot{q}_v = -\frac{\partial}{\partial q_v} V(\vec{q}) = F_v$! ~ Newton's 2nd Law

Or in vec. terms:

$$\frac{d}{dt}\frac{\partial L}{\partial \vec{\dot{q}}} - \frac{\partial L}{\partial \vec{q}} = \frac{d}{dt} m\vec{\dot{q}} - \vec{\nabla}(-V) \Rightarrow m\vec{\ddot{q}} = -\vec{\nabla} V$$

$$\sim \left(m\ddot{q}_x, m\ddot{q}_y, m\ddot{q}_z\right) = -\left(\frac{\partial V}{\partial q_x}, \frac{\partial V}{\partial q_y}, \frac{\partial V}{\partial q_z}\right) = -\left(\frac{\partial}{\partial q_x}, \frac{\partial}{\partial q_y}, \frac{\partial}{\partial q_z}\right) V = -\vec{\nabla} V$$

→ Thus recovering the movement equation from Newtonian Mechanics!

• Expressing the E.L. equations for 3 coordinates (and using $\vec{\dot{q}} = \vec{v}$):

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \vec{v}}\right) = \vec{\nabla} L}$$

→ If $L = T - V$: $\frac{d}{dt}(m\vec{v}) = \vec{\nabla}(-V) \Rightarrow \frac{d\vec{p}}{dt} = -\vec{\nabla} V \Rightarrow \frac{d\vec{p}}{dt} = \vec{F}$

Newton's 2nd Law very directly!

Now that this is clear, we can go back to the wake-fields context and find the Lagrangian of the witness particle (in whose $\Delta\vec{p}$ - "α" $\vec{w}$ - we're interested in).

It is the Lagrangian of a <u>relativistic charged particle</u> moving through E.M. fields $\vec{E}$ and $\vec{B}$, i.e., under the influence of <u>E.M. potentials ϕ and $\vec{A}$</u>.

Let's build it in two parts:

- Relativistic free-particle term
- Electromagnetic interaction term

APPENDIX

(i) Lagrangian of a Relativistic Free-Particle

• In non-relativistic mechanics, we would have simply: $L = T = \frac{1}{2}mv^2$

• But in the relativistic context, we also have the requirement of $S = \int L \cdot dt$ being Lorentz invariant! ~ which is not the case for dt, ds or v ($\neq$ observers can disagree about them).

→ Thus, we should have something with the units of action (energy · time) which is also Lorentz invariant.

↳ The natural invariant along a particle's path is its proper time $d\tau$ → actual time experienced by the particle → independent of the observer's ref. frame → fixed by the geom. of spacetime

$$d\tau = \frac{dt}{\gamma}$$

dt → time interval measured by a stationary observer's reference frame

$d\tau$ → time registered by a clock which is $\underbrace{\text{at rest relative}}_{\text{moving with}}$ to the particle

→ Then, to adjust for the units, the simplest L.-inv. term with energy units would be $A \cdot mc^2$ ~ prop. to rest mass:

$$\Rightarrow S = \int A \cdot mc^2 \cdot d\tau = \int A \cdot \frac{mc^2}{\gamma} dt$$

$L = A mc^2 \sqrt{1 - v^2/c^2}$

Finally, to determ. A we can use Lagrangian formalism, from which the canonical momentum is:

$$\vec{P}_{can} = \frac{\partial L}{\partial \vec{v}} = \frac{\partial}{\partial \vec{v}}\left(A mc^2 \sqrt{1 - v^2/c^2} \right) = A mc^2 \cdot \frac{-2v/c^2}{2\sqrt{1 - v^2/c^2}} = -A\gamma m\vec{v}$$

Thus, for it to match the relativistic momentum, $\vec{P}_{rel} = \gamma m \vec{v}$, we need $A = -1$

And therefore, for a rel. free-particle the Lagrangian is:

$$L_{rel} = -\frac{mc^2}{\gamma} = -mc^2\sqrt{1 - v^2/c^2}$$

~ Notice that if $v << c \Rightarrow \frac{v^2}{c^2} << 1$:

$$(1-u)^{1/2} = 1 + -\frac{1}{2}(1-u)^{-1/2}\Big|_{u=0} \cdot u + \ldots \overset{u<<1}{\cong} 1 - \frac{u}{2}$$

and thus: $L_{rel} \overset{v<<c}{\cong} -mc^2\left(1 - \frac{v^2}{2c^2}\right) = -mc^2 + \frac{1}{2}mv^2$ ✓

↳ Cte. doesn't affect the eq's of motion: $\Rightarrow \frac{\partial L}{\partial \vec{v}} = m\vec{v} = \vec{P}_{classical}$

APPENDIX

(ii) Electromagnetic Interaction Lagrangian Term

- Forgetting relativity for a moment $(v \ll c)$, we want to write a Lagrangian for which the Euler-Lagrange eq. gives a force term which is precisely the Lorentz Force, $\vec{F}_L = q(\vec{E} + \vec{v} \times \vec{B})$.

- We remember that $\vec{E}$ and $\vec{B}$ (due to satisfying $\vec{\nabla} \times \vec{E} = 0$ and $\vec{\nabla} \cdot \vec{B} = 0$) can be written in terms of the electric and magnetic potentials: $\vec{E} = -\vec{\nabla}\phi$, $\vec{B} = \vec{\nabla} \times \vec{A}$.

  $\sim$ $\phi = \phi(\vec{r})$ ~ scal. 3D func (ϕ: $3 \to 1$); $\vec{A} = \vec{A}(\vec{r})$ ~ vec. 3D field ($\vec{A}$: $3 \to 3$)

  $\overset{\cdot q}{\Rightarrow} \vec{F}_E = -\vec{\nabla} V_{el}$ ~ *electrostatic case*

- To build an intuition, we start with the case where only an electric force is acting on the particle (purely electrostatic ~ $\vec{B} = 0 \Rightarrow \vec{E} = -\vec{\nabla}\phi$)

  → The potential energy is then: $V_{el} = q \cdot \phi(\vec{r})$

  → Plugging into the Lagrangian: $L = \frac{1}{2} m |\vec{v}|^2 - q\phi$

  → Euler-Lagrange: $\frac{d}{dt}\left(\frac{\partial L}{\partial \vec{v}}\right) = \vec{\nabla} V$

  $\Rightarrow \frac{d}{dt}\vec{p} = -\vec{\nabla} \cdot q\phi \Rightarrow \vec{F}_E = -\vec{\nabla} V_{el}$ ✓

- From now on we should make the Maxwell correction $\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$

- Now, to account for the $q\vec{v} \times \vec{B}$ of the Lorentz force, we see that

$$\vec{v} \times \vec{B} = \vec{v} \times \vec{\nabla} \times \vec{A} = \vec{\nabla}(\vec{v} \cdot \vec{A}) - \vec{A}(\vec{v} \cdot \vec{\nabla}),$$

B(A·C) − C(A·B) rule w/ operator awareness for 2nd term

and based on the 1st term, make the guess:

→ $V = q\phi - q\vec{v} \cdot \vec{A}$

→ Lagrangian: $L = \frac{1}{2} m |\vec{v}|^2 - q\phi + q\vec{v} \cdot \vec{A}_{(\vec{r})}$

→ E.L.: $\frac{d}{dt}\left(\frac{\partial L}{\partial \vec{v}}\right) - \vec{\nabla} L = \frac{d}{dt}\left(m\vec{v} + q \cdot \vec{A}\right) - q\vec{\nabla}\left(-\phi + \vec{v} \cdot \vec{A}\right) =$

$$= \frac{d\vec{p}}{dt} + q\frac{d\vec{A}}{dt} + q\vec{\nabla}\phi - q\vec{\nabla}(\vec{v} \cdot \vec{A}) = 0 \quad (*)$$

But notice that $\frac{d}{dt}\vec{A}(\vec{r}) = \frac{\partial \vec{A}}{\partial t} + \frac{\partial \vec{r}}{\partial t} \cdot \frac{\partial}{\partial \vec{r}}\vec{A} = \frac{\partial \vec{A}}{\partial t} + (\vec{v} \cdot \vec{\nabla})\vec{A}$

$\sim \frac{dA_x(\vec{r},t)}{dt} = \frac{\partial A_x}{\partial t} + \underbrace{\frac{\partial A_x}{\partial x} \cdot \frac{\partial x}{\partial t} + \dots + \frac{\partial A_x}{\partial z}\frac{\partial z}{\partial t}}_{= (\vec{\nabla} \cdot A_x) \cdot \vec{v}} = \frac{\partial A_x}{\partial t} + (\vec{v} \cdot \vec{\nabla}) A_x$

$\left(\frac{\partial x}{\partial t}\frac{\partial}{\partial x} + \frac{\partial y}{\partial t}\frac{\partial}{\partial y} + \frac{\partial z}{\partial t}\frac{\partial}{\partial z}\right)(A_x, A_y, A_z)$

And thus:

$$(*) \Rightarrow \frac{d\vec{p}}{dt} = +q\left[-\frac{\partial \vec{A}}{\partial t} \underbrace{- \overbrace{(\vec{v} \cdot \vec{\nabla})\vec{A}}^{([\vec{\nabla} \cdot A_x] \cdot \vec{v}, \dots)} + \vec{\nabla}(\vec{v} \cdot \vec{A})}_{= \vec{v} \times (\vec{\nabla} \times \vec{A})!} - \vec{\nabla}\phi\right] = q\left[\underbrace{-\left(\vec{\nabla}\phi + \frac{\partial \vec{A}}{\partial t}\right)}_{\vec{E}} + \underbrace{\vec{v} \times (\vec{\nabla} \times \vec{A})}_{\vec{v} \times \vec{B}}\right]$$

$$\Rightarrow \frac{d\vec{p}}{dt} = q(\vec{E} + \vec{v} \times \vec{B}) = F_L \quad //$$

~ properly giving Newton's 2nd Law for a non-relativistic charged particle w/ velocity $\vec{v}$ under effect of the Lorentz force. ✓